



Serpentine

△ **Serpentine specimen** from Lowell, Vermont, USA

Serpentine is not just one mineral, but a group of at least 16 white, yellowish, green, or grey-green magnesium minerals with a complex chemistry and similar appearance. Serpentine minerals generally occur as masses of tiny, intergrown crystals, and are named in allusion to their mottled appearance, which resembles a snake's skin. Gem-quality serpentine, often with a jade-like appearance, is cut *en cabochon*. Soft enough to be engraved, the mineral is also used in carvings. It is widespread, and there are huge quarrying operations in various parts of the world.

Specification

Chemical name Magnesium, iron, or nickel silicate | **Formula** $(\text{Mg,Fe,Ni})_3\text{Si}_2\text{O}_5(\text{OH})_4$ | **Colours** White, grey, yellow, green, green-blue | **Structure** Monoclinic | **Hardness** 2.5–5.5 | **SG** 2.5–2.6 | **RI** 1.56–1.57 | **Lustre** Subvitreous to greasy, resinous, earthy, dull | **Streak** White | **Locations** Worldwide

High-grade serpentine | **Rough** | The translucency of this example of fine green serpentine shows how it can easily be mistaken for jade.

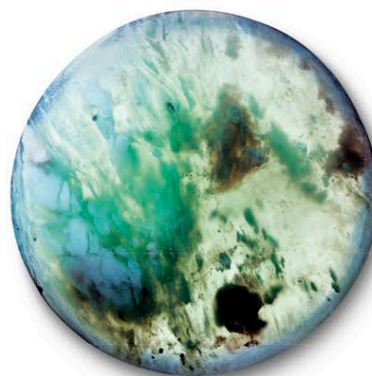


Internal fractures

Rock groundmass



Serpentine with white chrysotile | **Rough** | Chrysotile is one of the serpentine minerals, and is one of a group of minerals commonly called "asbestos".



Williamsite cabochon | **Cut** | Williamsite is one of the serpentine varieties used as gems, and produces interesting cabochons, as seen in this example.



Neolithic carving | **Carved** | Mysterious serpentine carvings such as this are found in archaeological sites across northern Britain, and date from as long as 4,500 years ago.



Seashell carving | **Colour variety** | Serpentine can come in many colours, such as the light green serpentine of this finely executed carving of a seashell.

Serpentines quarried as ornamental stones are sometimes called serpentine marble



Bowenite pendant | **Carved** | Bowenite is a variety of antigorite, a type of serpentine. Like much serpentine, it can be carved in fine detail, as demonstrated here.